Local Obstructions and Slowly Growing Search Depth in the Erdős–Straus Conjecture: A Classified Census to 10^{15}

Vincent*

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Abstract

For a prime p, write $A = (p + r)/4$ with $r \equiv 3 \pmod 4$ and $q = pA$. The Erdős–Straus equation has a solution at this A exactly when a divisor of q^2 lies in the class $-q \pmod r$. This turns the search into a ladder over the rungs $r = 3, 7, 11, \ldots$ and separates failed rungs into four levels: obstruction by the Jacobi symbol, by another real character, by the multiplicative support subgroup, or a *residual miss* for which the target class is locally available but no divisor reaches it.

We classify every failed rung for the 932,642,160,749 primes below 10^{15} in the six square classes modulo 840 not covered by Mordell's identities. Among 558 billion failed rungs, 99.1773% have a Jacobi certificate, 0.00008% require a finer local certificate, and 0.8227% are residual. The finest local level occurs only at the single rung $r = 51$, in all 43,257 instances, and we prove that it must: since $4A = p + r$, the support subgroup always contains 4—a constraint invisible to every real character, because $\chi(4) = \chi(2)^2 = 1$, but decisive for the support test. It confines that level to ten rungs below 300, of which 51 is the smallest. The residual share falls monotonically across the census, but its absolute count continues to grow. The maximum first-hit rung also rises, from the long-standing record 107 through 131 to 155. The data therefore gives no support for a uniform constant bound on search depth; instead it suggests an unbounded, exceptionally slowly growing envelope.

The residual misses are concentrated among divisor-poor shifted values $(p+r)/4$. However, a random-occupancy model for the divisors of q^2 fails by orders of magnitude, showing that multiplicative dependence, not merely divisor density, is the central obstacle. These results are numerical and do not prove a new case of the conjecture. Their contribution is a reproducible decomposition of what local residue information explains and what it leaves arithmetically unresolved.

1 Introduction

The Erdős–Straus conjecture [2] asserts that for every integer $n \geq 2$ the equation $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers; it suffices to treat prime $n = p$. Polynomial congruence identities resolve the equation outside a thin set of residue classes. Mordell's

*Computations performed with machine assistance; all numerical claims are reproducible from the released code and data (§9). Code, data, and discussion are hosted at `https://github.com/Attitude-Adjuster/erdos-strauss`; questions, corrections, and reproducibility reports may be filed through the repository's issue tracker.

identities leave six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$, all of them squares modulo 840 [8]. We denote their union by the *hard class* $\mathcal{H}$. The common surrogate $p \equiv 1 \pmod{24}$ is four times larger and mixes these primes with classes already settled by an identity; Section 4.1 shows that this dilutes the measured residual share. We therefore census $\mathcal{H}$ directly. The obstruction is structural: Schinzel and Yamamoto showed that no finite list of polynomial congruence identities covers a quadratic-residue class [10, 15].

Fixing A and setting $r = 4A - p$, $q = pA$, solvability is equivalent to the existence of a divisor of q^2 in the class $-q \bmod r$ (Lemma 2.1, classical). Failure can then have two very different causes:

- *Local residue support.* The divisors of q^2 all lie in some proper subset of $(\mathbb{Z}/r)^\times$ determined by the residues of the primes of q, and the target class lies outside it. No search is required; a short certificate settles it.
- *A residual miss.* The target class is available to the divisors in every local sense, but no divisor happens to land there.

The first is finite group arithmetic. The second is the unresolved part of the search. We make this separation explicit through a nested certificate hierarchy, classify it exhaustively on $\mathcal{H}$ below 10^{15}, and release independently checkable certificates. The parametrization, character expansion, and Type I/II taxonomy are classical; the contribution here is the hierarchy, a structural theorem pinning its finest local level to an explicit finite list of rungs (§3.1), the exhaustive classification, and what it reveals about the slowly growing residual tail.

2 The rung ladder and the divisor lattice

Fix a prime $p \equiv 1 \pmod 4$ and an integer $A > p/4$, and set $r = 4A - p > 0$, $q = pA$. Subtracting $1/A$ from $4/p$ gives $1/B + 1/C = r/q$, and completing the product turns this into a hyperbola:

$$\frac{4}{p} = \frac{1}{A} + \frac{1}{B} + \frac{1}{C} \qquad \Longleftrightarrow \qquad (rB - q)(rC - q) \;=\; q^2. \tag{1}$$

Lemma 2.1 (divisor criterion; classical)**.** *Positive integer solutions* (B, C) *of* (1) *correspond bijectively to divisors* $u \mid q^2$ *with* $u \equiv -q$ *and* $q^2/u \equiv -q \pmod r$*, via* $B = (u+q)/r$*,* $C = (q^2/u + q)/r$*. If* $\gcd(q, r) = 1$ *the second congruence follows from the first.*

Proof. If (B, C) solves (1) then $u = rB - q$ divides q^2 and is $\equiv -q \pmod r$, as is its cofactor; conversely such a u yields integers $B, C \ge 1$. When $\gcd(q, r) = 1$, $u \equiv -q$ gives $q^2/u \equiv q^2(-q)^{-1} \equiv -q \pmod r$. □

As A runs over integers exceeding $p/4$, r runs over $r \equiv -p \equiv 3 \pmod 4$. We call the resulting search order a *ladder* with *rungs* $r = 3, 7, 11, \ldots$ and $A = (p + r)/4$; rung r *hits* if Lemma 2.1 produces a solution, and the *depth* of p is the smallest hitting r.

The required coprimality is automatic throughout the range of interest.

Lemma 2.2. *If* $0 < r < 3p$ *then* $\gcd(q, r) = 1$.

Proof. $r < 3p$ is equivalent to $A < p$, so $p \nmid A$ and, p being prime, $\gcd(A, p) = 1$. Then $\gcd(A, r) = \gcd(A, 4A - p) = \gcd(A, p) = 1$ and $\gcd(p, r) = \gcd(p, 4A - p) = \gcd(p, 4A) = 1$ since p is odd and $p \nmid A$. Hence $\gcd(pA, r) = 1$. □

Thus the second congruence in Lemma 2.1 is not an additional condition in the census.

The divisors of $q^2 = p^2A^2$ form a box lattice: writing $A = p^eA'$ with $p \nmid A'$ and $A' = \prod_i \ell_i^{a_i}$, they are $u = p^jd$ with $0 \le j \le 2e+2$ and $d \mid A'^2$. Reducing modulo r colors this grid, and Lemma 2.1 asks whether the color $-q$ appears. The classical two-type taxonomy is the row coordinate:

Proposition 2.3 (types are rows)**.** *Suppose $p \nmid A$ and $\gcd(q,r) = 1$, and let a solution arise from $u = p^jd$ with $j \in \{0,1,2\}$ and $d \mid A^2$. If $j = 1$ then $p \mid B$ and $p \mid C$ (Type II); if $j \in \{0,2\}$ then p divides exactly one of B, C (Type I).*

Proof. $p \mid q$, so $p \mid u+q$ iff $p \mid u$, i.e. iff $j \ge 1$; the cofactor has p-exponent $2-j$. Since $\gcd(r,p) = 1$, $p \mid B$ iff $p \mid u+q$. □

Remark 2.4. For $p = 73$, $A = 20$ ($r = 7$, $q = 1460$) the lattice contains four solution divisors $u \le q$: $u = 10, 80$ in row $j = 0$ (Type I, e.g. $4/73 = 1/20 + 1/210 + 1/30660$) and $u = 73, 584$ in row $j = 1$ (Type II, e.g. $4/73 = 1/20 + 1/219 + 1/4380$). Both classical families sit in one lattice.

3 The character expansion and a hierarchy of certificates

Throughout, $\gcd(q,r) = 1$ and $N_r(p) = \#\{u \mid q^2 : u \equiv -q \pmod{r}\}$, counting all divisors (each unordered solution $\{B, C\}$ contributes two).

Theorem 3.1 (divisor count in a class; orthogonality)**.** *With $q^2 = \prod_\ell \ell^{a_\ell}$,*

$$N_r(p) \;=\; \frac{1}{\varphi(r)} \sum_{\chi \pmod{r}} \overline{\chi(-q)} \prod_{\ell \mid q} \bigl(1 + \chi(\ell) + \chi(\ell)^2 + \cdots + \chi(\ell)^{a_\ell}\bigr). \tag{2}$$

Proof. For u, a coprime to r, orthogonality [1, Ch. 4] gives $\varphi(r)^{-1} \sum_\chi \chi(u)\overline{\chi(a)} \;=\; \mathbf{1}[u \equiv a]$. Take $a = -q$, sum over $u \mid q^2$, and factor the completely multiplicative χ across the divisor structure of q^2. □

The conjugate in (2) is essential: omitting it counts the class $(-q)^{-1}$ rather than $-q$. Table 3 checks both evaluations against direct enumeration.

The trivial character contributes $\tau(q^2)/\varphi(r) > 0$; the others contribute cancellation. For real characters that cancellation is itself a divisor count:

Lemma 3.2. *Let χ be a real character mod r. Since every exponent a_ℓ in q^2 is even,*

$$\sum_{u \mid q^2} \chi(u) \;=\; \prod_{\chi(\ell)=1} (a_\ell + 1) \;=\; \tau\bigl((q^2)_+\bigr) \;\ge\; 0,$$

where $(q^2)_+$ is the part of q^2 supported on primes with $\chi(\ell) = 1$: at primes with $\chi(\ell) = -1$ the local factor $1 - 1 + \cdots + 1$ equals 1.

The obstruction applies to every odd r and every real character.

Corollary 3.3 (real-character certificate)**.** *Let r be odd with $\gcd(q,r) = 1$ and let χ be any real Dirichlet character mod r. If $\chi(\ell) = +1$ for every prime $\ell \mid q$ and $\chi(-q) = -1$, then $N_r(p) = 0$.*

Proof. χ is completely multiplicative, so every divisor $u \mid q^2$ has $\chi(u) = +1$, whereas χ is constant on residue classes mod r and equals -1 at $-q$. Hence no divisor lies in that class. Equivalently in (2): by Lemma 3.2 the χ term is $\chi(-q)\tau(q^2) = -\tau(q^2)$, cancelling the trivial term exactly. $\square$

Remark 3.4 (the Jacobi symbol is only one of them). For odd $r = \prod_{i=1}^{\omega} p_i^{e_i}$ the group $(\mathbb{Z}/r)^\times$ has exactly 2^ω real characters, namely the products $\chi_T(x) = \prod_{i\in T} \left(\frac{x}{p_i}\right)$ over subsets $T \subseteq \{1, \dots, \omega\}$. The Jacobi symbol is one of them—but *not* in general the full-subset one: $\left(\frac{\cdot}{r}\right) = \chi_{T_0}$ with $T_0 = \{i : e_i \text{ odd}\}$, since $\left(\frac{x}{p_i^{e_i}}\right) = \left(\frac{x}{p_i}\right)^{e_i}$. For squarefree r these coincide; for $r = 63 = 3^2 \cdot 7$ the Jacobi symbol is $\left(\frac{\cdot}{7}\right)$ and for $r = 75 = 3 \cdot 5^2$ it is $\left(\frac{\cdot}{3}\right)$, and both of these rungs occur in our data. For prime r there is a single nontrivial real character and the distinction is vacuous; for composite r, testing only the Jacobi symbol is strictly weaker than testing all real characters. Note also that $\left(\frac{x}{r}\right) = +1$ does not mean x is a quadratic residue when r is composite, so the hypothesis must be phrased in terms of the character, not of residuosity.

The criteria simplify considerably once one uses $r \equiv 3 \pmod 4$, and the simplification explains exactly which rungs can exhibit which levels.

Proposition 3.5 (criteria in reduced form)**.** *Let rung r fail for p, and let $H \le (\mathbb{Z}/r)^\times$ be as in Definition 3.6. Recall $r \equiv 3 \pmod 4$.*

1. *A real character χ mod r certifies the failure if and only if $\chi(-1) = -1$ (χ is odd) and $H \le \ker\chi$. In particular the Jacobi symbol certifies if and only if $\left(\frac{\ell}{r}\right) = +1$ for every prime $\ell \mid q$: the hypothesis $\left(\frac{-q}{r}\right) = -1$ is automatic.*
2. *$-q \in H$ if and only if $-1 \in H$. Hence the support certificate of Definition 3.6 says exactly that $-1 \notin H$.*
3. *There are exactly $2^{\omega(r)-1}$ odd real characters mod r, one of which is the Jacobi symbol. Consequently a level-*RC *failure can occur only when $\omega(r) \ge 2$.*

Proof. Since q is a product of the primes generating H, we have $q \in H$; so if $H \le \ker\chi$ then $\chi(q) = 1$ and $\chi(-q) = \chi(-1)$, giving (1). For the Jacobi symbol, $\left(\frac{-1}{r}\right) = (-1)^{(r-1)/2} = -1$ because $r \equiv 3 \pmod 4$. Part (2) is the same observation: $-q \in H \iff -1 \in H$ because $q \in H$. For (3), $\chi_T(-1) = \prod_{i\in T}(-1)^{(p_i-1)/2}$, so χ_T is odd exactly when T meets $\{i : p_i \equiv 3 \pmod 4\}$ in an odd number of elements; that set is nonempty since $r \equiv 3 \pmod 4$, and the number of such T is $2^{\omega-1}$. When $\omega = 1$ there is therefore exactly one odd real character, which must be the Jacobi symbol, leaving no room for level RC. $\square$

Definition 3.6 (certificate hierarchy)**.** Let rung r fail for p (so $\gcd(q, r) = 1$ by Lemma 2.2), and let $H \le (\mathbb{Z}/r)^\times$ be the subgroup generated by $\{\ell \bmod r : \ell \mid q\}$. The failure has level

J if the Jacobi symbol $\left(\frac{\cdot}{r}\right)$ satisfies Corollary 3.3;

RC if some real character mod r does, but the Jacobi symbol does not;

S if no real character does, but $-q \bmod r \notin H$;

R (*residual miss*) otherwise: $-q \in H$, yet the divisors of q^2—whose exponents are bounded—miss the class.

Proposition 3.7. *The levels are nested. Every J-certificate is an RC-certificate, and every RC-certificate implies $-q \notin H$. Level R is what remains after all local support information is exhausted.*

Proof. The first claim is Remark 3.4. For the second, if χ is trivial on every $\ell \mid q$ then $H \leq \ker \chi$, while $\chi(-q) = -1$ puts $-q$ outside $\ker \chi$. □

Levels RC and S are finer tests in principle—S detects obstructions from higher-order characters and from subgroups of index greater than 2 that no real character sees. Whether they are finer *in practice*, and at what height they first bite, is an empirical question answered in §4; §3.1 shows that level S is in addition confined to an explicit finite set of rungs.

For the smallest rung, level R is empty unconditionally, because modulo 3 there are only two characters.

Proposition 3.8 (rung 3 has no residual misses)**.** *Let $p \equiv 1 \pmod{12}$ and $A = (p+3)/4$, so $r = 3$. Then rung 3 hits unless it carries a real-character certificate; explicitly, it fails precisely when $3 \nmid A$ and every prime $\ell \mid pA$ satisfies $\ell \equiv 1 \pmod 3$.*

Proof. If $3 \mid A$ take $u = 3$: then $3 \mid q$, $9 \mid q^2$, and both congruences of Lemma 2.1 read $0 \equiv 0 \pmod 3$, so rung 3 hits. If $3 \nmid A$ then $\gcd(q, 3) = 1$ and (2) has two terms, $N_3(p) = \frac{1}{2}\big(\tau(q^2) + \chi(-q)\,\tau((q^2)_+)\big)$ with χ the nontrivial character mod 3 (real, so conjugation is immaterial). If some $\ell \mid q$ has $\chi(\ell) = -1$ then $\tau((q^2)_+) < \tau(q^2)$ and $N_3 > 0$ either way. If every $\ell \mid q$ has $\chi(\ell) = 1$ then $\tau((q^2)_+) = \tau(q^2)$ and $\chi(-q) = \chi(-1) = -1$, so $N_3 = 0$; since $\big(\frac{\cdot}{3}\big)$ is the only nontrivial real character mod 3, this is a level-J certificate. □

3.1 Which rungs can carry a level-S failure

Level S is the one level no real-character certificate detects, and empirically (§4) it sits at a single rung. Higher-order characters still detect the support obstruction, as the centered Fourier formulation in §7 makes explicit. Its confinement is forced by a constraint on H that every real character is blind to.

Lemma 3.9 (the four-constraint)**.** *$4 \in H$, always.*

Proof. $4A = p + r$ gives $A \equiv 4^{-1}p \pmod r$, the inverse existing because r is odd. Now H contains p and every prime factor of A, hence contains A itself; therefore $4^{-1} = A\,p^{-1} \in H$ and so $4 \in H$. □

Any real χ has $\chi(4) = \chi(2)^2 = 1$, so $4 \in \ker \chi$ automatically and Lemma 3.9 excludes nothing in Corollary 3.3. For the support test it is decisive.

Proposition 3.10 (when level S is possible)**.** *Write $G = (\mathbb{Z}/r)^\times$. Rung r can carry a level-S failure only if some subgroup $H \leq G$ satisfies*

$$4 \in H, \qquad -1 \notin H, \qquad (-1)H \in (G/H)^2, \tag{3}$$

and any H satisfying (3) does produce one whenever it is realized as a support subgroup. In particular level S is impossible at every prime rung.

Proof. The first condition is Lemma 3.9, the second is Proposition 3.5(2), and the second also forces the rung to fail, since all divisors of q^2 lie in H. For the third: the real characters trivial on H are exactly the characters of G/H of order dividing 2, and by Proposition 3.5(1) the failure escapes every real-character certificate precisely when all of them take the value $+1$ at -1. The intersection of the kernels of all order-≤ 2 characters of a finite abelian group is its subgroup of squares, which is the stated condition.

If r is prime then G is cyclic of order $r - 1 \equiv 2 \pmod 4$. Then $-1 \notin H$ forces $|H|$ odd, so G/H is cyclic of order $2m$ with m odd and $(-1)H$ is its unique element of order 2 — not a square, since $4 \nmid |G/H|$. No H qualifies. $\square$

Corollary 3.11. *Among $r \equiv 3 \pmod 4$ with $r < 300$, level S is possible only for*

$$r \in \{51,\ 119,\ 123,\ 187,\ 195,\ 219,\ 255,\ 267,\ 287,\ 291\},$$

and 51 is the smallest by a wide margin. At $r = 51$, 119 and 123 the admissible subgroup is unique and cyclic of index 4; at $r = 51$ it is $H = \langle 2 \rangle = \{1, 2, 4, 8, 13, 16, 26, 32\}$ inside $(\mathbb{Z}/51)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/16$.

Dropping Lemma 3.9 from (3) would admit fourteen rungs below 140 alone,

$$15,\ 35,\ 39,\ 51,\ 55,\ 75,\ 87,\ 91,\ 95,\ 111,\ 115,\ 119,\ 123,\ 135,$$

with nothing to distinguish 51, and with $r = 15$ visited far more often than $r = 51$. The four-constraint eliminates eleven of the fourteen. It also disposes of the objection that the confinement might be an artifact of which rungs are reached: level RC is observed at $r = 51, 75, 91, 99, 115$, so those rungs *are* reached carrying local obstructions, yet at 75, 91 and 115 every subgroup that would otherwise satisfy (3) omits 4, and at 99 none exists in the first place.

The statement is falsifiable at fixed cost: **no census, at any height, may report a level-S failure at** $r = 15, 35, 39, 55, 75, 87, 91, 95, 111, 115$ **or** 135, and the next rung that can host one is $r = 119$.

4 Exhaustive classification below 10^{15}

The census follows every prime in $\mathcal{H}$ to its first hitting rung and classifies all preceding failures by Definition 3.6. It contains 932,642,160,749 primes and 557,964,077,963 failed rungs below 10^{15}; the 10^{13} stage was produced on CPUs and the extension to 10^{15} by a GPU implementation whose statistics and certificate set are byte-identical to the CPU reference on every range where both run. Independent exact-arithmetic and machine-word implementations agree on all overlapping ranges, and every emitted fine-certificate or deep-prime claim—all 585,677 of them—was checked again in exact arithmetic on a machine different from the one that produced it. Section 5 gives the algorithm and verification protocol.

The depth record remained 107 across three decades of new primes, then moved to 111, to 131, and—within the fifteenth decade—to 155, at

$$p = 172{,}538{,}390{,}619{,}841, \qquad \frac{4}{p} = \frac{1}{43134597654999} + \frac{1}{48015316516502146586034030} + \frac{1}{5672750915411526282158020334269004\,70},$$

Table 1: Observed maximum search depth $D(p)$, the first hitting rung, on $\mathcal{H}$. Long plateaus are followed by small record increments; they are evidence of an extreme tail, not of a fixed ceiling.

cutoff X	$\max_{p<X} D(p)$	first record holder in the census
10^8	107	8,803,369
10^{10}	107	8,803,369
10^{11}	107	8,803,369
10^{12}	111	654,730,707,409
10^{13}	131	6,367,024,012,441
10^{14}	131	6,367,024,012,441
10^{15}	155	172,538,390,619,841

and the displayed decomposition is exact. Below 10^{15} the depths 111, 115, 119, 123, 127, 131 (eight primes), 135 (three), and 139 (one) are all occupied, while 143, 147, and 151 are not: the record sits beyond a gap, as 107 once did. The long plateaus are therefore sampling effects in an exceptionally thin tail. Numerically, the natural expectation is not a constant bound $D(p) \leq R$, but an unbounded envelope that grows too slowly to estimate reliably from record values alone. This is suggestive, not a theorem; the current record is no more a bound than its predecessor was.

All failures at $r = 3$ are level J, as Proposition 3.8 predicts. Residual misses first appear at $r = 11$ and are concentrated at small rungs, while the finer local levels occur only on the thin subsequence of composite rungs. Their full counts and localization are reported below on the same population as the main census.

4.1 Which primes are actually hard

Mordell's identities [8] settle $4/n$ for every n except possibly those n that are *squares* modulo 840. There are exactly six such classes,

$$\mathcal{H} \;=\; \{1,\ 121,\ 169,\ 289,\ 361,\ 529\} \pmod{840}, \tag{4}$$

being $1^2, 11^2, 13^2, 17^2, 19^2, 23^2$. All six lie in 1 (mod 24), but that coarse class contains 24 reduced classes modulo 840, only six of which are squares. The other three quarters have already been settled, yet their early failed rungs would still enter the denominator of any reported failure share.

Observation 4.1 (the easy primes contribute only level J). Through 10^{11}, the censuses of (1 mod 24) and $\mathcal{H}$ have identical depth counts at every rung $r \geq 11$ and identical level-RC, level-S, and level-R totals. Every extra failure from (1 mod 24) $\setminus \mathcal{H}$ occurs at $r = 3$ and has a Jacobi certificate. Restricting to $\mathcal{H}$ therefore removes only already-explained failures and preserves the entire residual and deep-ladder population in the overlap.

Including the coarse classes thus understates the residual share. All subsequent depth and failure-share statistics use $\mathcal{H}$.

Observation 4.2 (the levels on $\mathcal{H}$). The classified census on the hard class is:

	primes	level J	level RC	level S	level R	residual share
$p < 10^6$	2,370	3,318	2	0	112	3.2634%
$p < 10^7$	20,513	24,376	2	0	662	2.6438%
$p < 10^8$	179,468	186,590	4	2	4,018	2.1079%
$p < 10^9$	1,587,581	1,473,032	18	4	26,317	1.7552%
$p < 10^{10}$	14,215,707	11,942,520	87	8	181,523	1.4972%
$p < 10^{11}$	128,671,219	99,140,159	454	47	1,302,101	1.2964%
$p < 10^{12}$	1,175,215,396	838,863,859	2,279	242	9,664,186	1.1389%
$p < 10^{13}$	10,814,395,686	7,211,448,416	12,112	1,309	73,736,922	1.0121%
$p < 10^{14}$	100,154,081,439	62,821,444,770	66,767	7,373	576,087,976	0.9087%
$p < 10^{15}$	932,642,160,749	553,373,502,288	386,596	43,257	4,590,145,833	0.8227%

Two features stand out.

- **Level S occurs only at** $r = 51$, across all 43,257 instances, while its sibling level RC spreads to $r = 51$ (382,218), 75 (4,137), 91 (169), 99 (59), 115 (7) and 123 (6). The rung grid does not force this localization — but Corollary 3.11 does: 51 is the smallest rung at which level S is possible at all, and 75, 91, 99, 115 are rungs at which it is impossible. Each of the 43,257 therefore has support subgroup $H = \langle 2 \bmod 51 \rangle$, the unique admissible one. The appearance of level RC at $r = 123$ sharpens the reachability point: 123 is on the admissible list of Corollary 3.11 and is now demonstrably reached carrying local obstructions, yet level S still declines to appear there — at $r = 123$ the admissible subgroup must arise as an actual support subgroup, which evidently remains rare.

- **The residual share falls monotonically**, from 3.26% to 0.8227%, while the residual count rises from 112 to 4,590,145,833. The proportions and the counts therefore tell different stories.

4.2 The central empirical thesis

We state the thesis the census supports, and—as carefully—the stronger statements it does not.

> *Local certificates asymptotically dominate the observed failure landscape; the remaining risk is concentrated in a sparse, divisor-poor tail.*

Formally we separate three hypotheses, which are often conflated and have very different status.

H1 (local dominance). Among failed rungs *of primes in* $\mathcal{H}$, the residual share tends to zero:

$$\Pr\big(\text{level R} \,\big|\, \text{failure},\ p \le X\big) \longrightarrow 0 \qquad (X \to \infty).$$

This is the hypothesis the data speaks to directly. *It does not imply the Erdős–Straus conjecture*: a vanishing share is compatible with a growing number of residual misses and says nothing about whether one prime can miss every rung.

H2 (divisor threshold). For each rung r there is a threshold $f(r)$ with $\tau(q^2) \ge f(r) \Rightarrow N_r(p) > 0$ whenever the target class is reachable. Residual misses still occur at $\tau(q^2) = 81$, so no small constant threshold is visible. A threshold depending on r remains open, and Section 4.4 shows why naive occupancy cannot establish one.

H3 (unbounded but slow depth). There is no uniform constant R with $D(p) \le R$ for every p, but the record envelope

$$M(X) = \max_{p<X,\ p\in\mathcal{H}} D(p)$$

grows exceptionally slowly. The move from 107 through 131 to 155 after long plateaus is consistent with this hypothesis and inconsistent with treating the old record as a natural barrier. Finite computation cannot distinguish genuine unboundedness from a larger constant bound, so H3 is deliberately stated as the numerical expectation, not as a conclusion.

4.3 Evidence for H1

The cumulative residual share on $\mathcal{H}$ is

$$3.26\%,\ 2.64\%,\ 2.11\%,\ 1.76\%,\ 1.50\%,\ 1.30\%,\ 1.14\%,\ 1.01\%,\ 0.9087\%,\ 0.8227\% \qquad (X = 10^6, \ldots, 10^{15}). \tag{5}$$

This decline is monotone, but cumulative censuses are nested and therefore unsuitable for a clean holdout test. Differencing them into disjoint decades gives

$[10^{k-1}, 10^k)$	$k = 7$	8	9	10	11	12	13	14	15
residual share	2.5454%	2.0269%	1.7038%	1.4608%	1.2688%	1.1178%	0.9954%	0.8953%	0.8116%

which is also monotone. A log-power fit to the first six decades predicts 0.9838% for $[10^{12}, 10^{13})$, against the observed 0.9954%, a 1.2% relative error on a disjoint range; the same family fitted through 10^{13} predicts 0.8844% and 0.7973% for the two decades that were computed only afterwards, against the observed 0.8953% and 0.8116% (1.2% and 1.8% relative error). This supports H1, but it does not establish a law or justify remote extrapolation. Most importantly, the falling share coexists with a rapidly increasing residual count.

4.4 Against random-divisor models: an occupancy test

There is a tempting heuristic for H2: if the $\tau(q^2)$ divisors were spread evenly over the $|H|$ reachable classes, the chance of missing one target class would be about e^{-m} with $m = \tau(q^2)/|H|$, so a large lattice would force a hit. We tested this directly on the recorded $p \equiv 1 \pmod{24}$ rung sample below 2×10^8, retaining only rungs whose target class is reachable (levels HIT and R), binning by m and comparing the observed miss rate to e^{-m} evaluated at the *lower* edge of each bin—the most generous value the null takes anywhere in the bin.

Table 2: Observed residual-miss rate against the random-occupancy null on the recorded $p \equiv 1$ (mod 24) sample, $p < 2 \times 10^8$.

$m = \tau(q^2)/\lvert H\rvert$	rungs	observed miss rate	$\max e^{-m}$ in bin	excess
$[1, 2)$	3,304	0.4837	3.7×10^{-1}	$1.3\times$
$[2, 4)$	5,543	0.4415	1.4×10^{-1}	$3.3\times$
$[4, 8)$	44,730	0.0194	1.8×10^{-2}	$1.1\times$
$[8, 16)$	397,224	3.8×10^{-3}	3.4×10^{-4}	$11\times$
$[16, 32)$	130,304	1.0×10^{-3}	1.1×10^{-7}	$9.2 \times 10^3\times$
$[32, 64)$	425,599	9.4×10^{-6}	1.3×10^{-14}	$7.1 \times 10^8\times$

The null is not merely inaccurate but wrong by growing orders of magnitude exactly where H2 would need it: at $m \in [16, 32)$ residual misses are nearly ten thousand times commoner than independence predicts, and beyond that the null predicts essentially none while misses still occur. Normalizing by $\varphi(r)$ instead of $|H|$ changes the table hardly at all, so this is *not* subgroup confinement—the support subgroup is usually everything. It is genuine dependence among the divisors of q^2 modulo r, which is unsurprising on reflection: those divisors are $\prod \ell_i^{f_i}$ with $f_i \leq 2e_i$, a rigid multiplicative lattice, not a random set.

The consequence for §7 is concrete: any route to H2 through equidistribution of divisors in residue classes—on average, or for typical integers—will not do, because the relevant configurations are precisely the atypical ones. This sharpens Direction III from "hard" to "hard for an identifiable reason."

4.5 How not to model the ladder

The rungs of one prime are not independent trials. They share p, while the shifted values $(p+3)/4, (p+7)/4, \ldots$ are consecutive integers with correlated factorizations. A statistical model must therefore treat the ladder as a rung-dependent survival process, not as repeated Bernoulli sampling. The observed shares are descriptive marginals; confidence intervals based on independence would be misleading.

Residual misses are nevertheless consistently divisor-poor: their median $\tau(q^2)$ is 27, compared with 81 at hitting rungs in the matched summaries. This is a correlation, not a consequence of (2); a small main term makes a zero possible, not probable.

The resulting decomposition of the covering problem is:

- a *local layer*, entirely finite: which rungs carry a real-character or support certificate is a computation in $(\mathbb{Z}/r)^\times$ given the residues of the primes of q;
- a *residual layer*: prove that level-R misses cannot occupy every rung of some prime. This is the content the conjecture actually needs, and H1 does not supply it—a vanishing share of failures is not a statement about any individual ladder. §4.4 shows the natural probabilistic route is blocked, and §7 that the natural analytic ones are too.

5 The computation

The census requires the complete factorization of every shifted value $A = (p+r)/4$ encountered before the first hit. Factoring these values independently is unnecessarily expensive; their arithmetic-progression structure permits a bulk sieve.

5.1 Bulk sieve-factorization

Within one of the six classes modulo 840, write $p = p_0 + 840j$. At a fixed rung,

$$A = \frac{p+r}{4} = A_0 + 210j, \qquad A_0 = \frac{p_0 + r}{4}. \tag{6}$$

For every prime $\ell \nmid 210$,

$$\ell \mid A \iff j \equiv -A_0\,\overline{210} \pmod{\ell}.$$

Thus each prime ℓ visits exactly the entries it divides. Sieving through $\sqrt{A_{\max}}$ leaves a prime cofactor, eliminating primality tests and Pollard rho from the main path. Once only a thin set of primes remains on the ladder, the scanner switches back to per-value factorization. Against the naive implementation this gives speedups of 7.3×, 12.8×, and 20.3× at 10^{10}, 10^{11}, and 10^{12}, respectively, with identical output.

5.2 Scaling

The measured per-prime cost grows slowly with height, so total cost is dominated by the number of primes. The range splits into independent shards and merges by adding histograms. On the machine used for the final run, the complete 10^{13} hard-class census took 17 minutes on 12 physical cores. The extension to 10^{15} was produced by a CUDA implementation of the same pipeline on a single NVIDIA L4 GPU in 2.6 hours ($\approx 9\,$ns per prime); its statistical output and its certificate set are byte-identical to the CPU reference on every range where both were run, which is the standard its adoption required. Different window sizes, shard decompositions, and the CPU/GPU implementation pair all produce identical statistics.

5.3 Verification protocol

The optimized scanners (CPU and GPU) are checked against an independent exact-arithmetic reference and against the earlier per-value implementation. Repeated runs, thread counts, window sizes, and independent shardings give identical counts. The final shards (192 at 10^{13}, 400 at 10^{15}) are checked for gaps and overlaps, and every fine-certificate and deep-prime record—20,001 at 10^{13}, 585,677 at 10^{15}—is re-derived in exact arithmetic on a machine different from the one that produced it. The verifier additionally enforces Proposition 3.10 mechanically: a level-S claim at an inadmissible rung, or one whose support subgroup omits 4, is rejected as falsifying rather than merely erroneous. Each reported solution is verified by the integer identity $4ABC = p(AB + BC + CA)$.

6 Relation to prior work

The divisor-congruence criterion of Lemma 2.1 is classical and appears in several equivalent guises. Mordell's identities [8] and the modular equations catalogued by Salez [9] are, in the present language, individual rungs: a polynomial identity valid on a residue class of p is a rule for choosing A (hence r) together with a divisor u that provably lands in the target class. Yamamoto [15] and Schinzel [10] established the quadratic obstruction that Corollary 3.3 localizes to a single rung. Elsholtz and Tao [3] give the Type I / Type II taxonomy used in Proposition 2.3, together with mean-value theorems for the number of solutions; their framework subsumes the parametrization here. Vaughan [14] bounded the exceptional set by $\ll N \exp(-c(\log N)^{2/3})$. The conjecture is verified numerically to 10^{18} [7], after Swett [11] and Salez [9]; a recent preprint [13] obtains density one within $n \equiv 1 \pmod 4$ using divisor conditions rather than congruence identities, and is the closest antecedent to the perspective taken here.

Accordingly we do *not* claim the divisor-lattice viewpoint as a new framework. Its ingredients are known; what we add is the certificate hierarchy of Definition 3.6, its exhaustive measurement, and the released certificates.

7 Directions: Fourier interference and its obstacles

The Fourier analogy becomes exact after centering the divisor lattice. Write $q = \prod_\ell \ell^{e_\ell}$ and let $G_r = (\mathbb{Z}/r)^\times$. A divisor $u = \prod_\ell \ell^{j_\ell}$ of q^2 determines, in G_r, the centered point $u/q = \prod_\ell \ell^{k_\ell}$ with $k_\ell = j_\ell - e_\ell$ and $-e_\ell \leq k_\ell \leq e_\ell$. Thus the moving target $-q$ becomes the fixed target -1.

Define the multiplicity measure

$$\nu_{q,r}(g) = \#\left\{ (k_\ell) : -e_\ell \le k_\ell \le e_\ell,\ \prod_{\ell | q} \ell^{k_\ell} = g \text{ in } G_r \right\}.$$

Then $N_r(p) = \nu_{q,r}(-1)$. With the convention $\widehat{\nu}(\chi) = \sum_{g \in G_r} \nu(g)\chi(g)$, and writing $\chi(\ell) = e^{i\theta_{\chi,\ell}}$, its finite Fourier transform factors as

$$\widehat{\nu}_{q,r}(\chi) = \prod_{\ell^{e_\ell} \| q} D_{e_\ell}(\theta_{\chi,\ell}), \qquad D_e(\theta) = \sum_{k=-e}^{e} e^{ik\theta} = \frac{\sin((2e+1)\theta/2)}{\sin(\theta/2)}, \tag{7}$$

where the quotient has the limiting value $2e+1$ at $\theta = 0$. Fourier inversion therefore rewrites Theorem 3.1 as

$$N_r(p) = \frac{1}{\varphi(r)} \left\{ \tau(q^2) + \sum_{\chi \neq \chi_0} \chi(-1) \prod_{\ell^{e_\ell} \| q} D_{e_\ell}(\theta_{\chi,\ell}) \right\}. \tag{8}$$

Every local factor is real, and $\chi(-1) = \pm 1$. A miss is consequently literal destructive interference: the nonprincipal spectrum cancels the positive mass $\tau(q^2)$ exactly.

This gives a deterministic, if not yet uniform, non-cancellation certificate. Put

$$\mathfrak{I}_r(p) = \sum_{\chi \neq \chi_0} \prod_{\ell^{e_\ell} \| q} \frac{|D_{e_\ell}(\theta_{\chi,\ell})|}{2e_\ell + 1}. \tag{9}$$

Proposition 7.1 (spectral non-cancellation)**.** *The triangle inequality in* (8) *gives*

$$N_r(p) \ \ge\ \frac{\tau(q^2)}{\varphi(r)} \big(1 - \mathfrak{I}_r(p)\big), \qquad \mathfrak{I}_r(p) < 1 \implies N_r(p) > 0. \tag{10}$$

Proof. Each normalized product in (9) is the absolute value of the corresponding nonprincipal term in (8), divided by $\tau(q^2)$. Subtract their sum from the principal term. □

Unlike a random-occupancy model, (10) is exact and retains the dependence among divisors. Its weakness is also precise: it discards the signs between nonprincipal modes.

The hierarchy now has a spectral reading. Let

$$B_{q,r} = \prod_{\ell^{e_\ell} \| q} \{\ell^{-e_\ell}, \ldots, 1, \ldots, \ell^{e_\ell}\} \subseteq G_r.$$

Then $N_r(p) > 0$ exactly when $-1 \in B_{q,r}$. A J- or RC-certificate is a maximally coherent odd real mode, trivial on every generator of the support subgroup H. At level S the same obstruction is detected by an odd higher-order character in the annihilator of H; it is invisible only if one restricts attention to real characters. At level R, $-1 \in H$ but the finite exponent box still misses it, so no annihilator character separates the target: cancellation is spread among partially damped modes. Equivalently, this is a product-set problem. Finite-group product-set estimates turn sufficiently small expansion of the successive factors in $B_{q,r}$ into a stabilizer subgroup, while expansion without a stabilizer is the combinatorial counterpart of Fourier damping. This structure-versus-mixing dichotomy is the main line of evidence behind the interference picture.

Direction I: spectral mixing on one rung. The immediate aim is to control (9) from the orders of the residues ℓ mod r and the exponents e_ℓ. If a nonprincipal character is nontrivial on several prime factors of q, the corresponding normalized Dirichlet kernels are damped. One needs enough such damping collectively to make $\mathfrak{I}_r(p) < 1$, or a signed refinement of (10) when the absolute-value bound is too expensive. Product-set growth offers a dual route: prove that $B_{q,r}$ either has a proper stabilizer, already recorded by the hierarchy, or covers -1 once its exponent box is sufficiently rich. This formulation explains why $\tau(q^2)$ alone cannot be a threshold: the phases and their multiplicities matter.

Direction II: decorrelation across the ladder. For $q_r = p(p+r)/4$, let $E_r(p)$ denote the nonprincipal part of (8), including the factor $1/\varphi(r)$, and set $\mathcal{R}_p(R) = \{r \le R : r \equiv 3 \pmod 4,\ \gcd(q_r, r) = 1\}$. Then

$$\sum_{r\in\mathcal{R}_p(R)} N_r(p) = \sum_{r\in\mathcal{R}_p(R)} \frac{\tau(q_r^2)}{\varphi(r)} + \sum_{r\in\mathcal{R}_p(R)} E_r(p). \tag{11}$$

(A noncoprime rung already hits by Lemma 2.2.) Hence total interference cannot persist through rung R if

$$\left|\sum_{r\in\mathcal{R}_p(R)} E_r(p)\right| < \sum_{r\in\mathcal{R}_p(R)} \frac{\tau(q_r^2)}{\varphi(r)}.$$

This is the rigorous form of the proposed "no total interference" mechanism. The difficulty is that both the group and the signal change with r: the signal at rung r is built from the factorization of q_r, so a standard large-sieve estimate does not apply unchanged. The sought-for statement is a cross-rung uncertainty principle for these arithmetically coupled spectra.

Direction III: arithmetic input for the moving signal. All analytic routes meet the same shifted-factorization obstacle. The Frobenius class of p controls $\chi(p)$, but not the prime factors of $(p+r)/4$. Selberg–Delange theory gives useful models for the frequency with which all those factors lie in a fixed subgroup—for an index-two condition, a count of order $c_r x(\log x)^{-1/2}$ follows from the usual branch-point argument [12, Ch. II.5]—but conditioning on p prime turns the question into a multiplicative correlation along a linear form. Chebotarev alone does not see that correlation, and sieve methods encounter the parity barrier.

For residual modes, the natural input is divisor distribution: Hooley's Δ-function [6], Hall–Tenenbaum [5], and Ford's work on divisors in prescribed ranges [4]. Those theories control typical integers; here one needs uniformity for the structured sequence $q_r^2 = p^2((p+r)/4)^2$, ultimately for every p. No known method makes that jump. The Fourier formulation does not remove this obstacle, but identifies the quantities that must be controlled: local phase damping on one rung or signed spectral decorrelation across several rungs.

8 Problems

Problem 1 (spectral census)**.** Augment the census with the normalized local factors in (9), the signed error $E_r(p)$, and the margin $1 - \mathfrak{I}_r(p)$. Determine how often (10) certifies a hit, which characters dominate residual misses, and whether every computed ladder contains a rung with $\mathfrak{I}_r(p) < 1$. For coherent J, RC and S modes, compare the observed frequencies with Selberg–Delange models that retain the coupling between p and $(p+r)/4$.

Problem 2 (structure versus mixing)**.** Prove a nontrivial condition on the orders of ℓ mod r and the exponents e_ℓ under which $-1 \in H$ forces $\mathfrak{I}_r(p) < 1$. Equivalently, give a product-set theorem forcing $-1 \in B_{q,r}$ once stabilizer obstructions have been excluded. Even a result for prime r, or for squarefree q, would separate genuine subgroup confinement from finite-box interference. A bound depending only on $\tau(q^2)$ cannot capture this distinction.

Problem 3 (cross-rung non-cancellation)**.** Find a range $R = R(p) < p$ on which the signed spectral errors in (11) satisfy

$$\left| \sum_{r \in \mathcal{R}_p(R)} E_r(p) \right| < \sum_{r \in \mathcal{R}_p(R)} \frac{\tau(q_r^2)}{\varphi(r)}.$$

Any such estimate for every $p \in \mathcal{H}$ proves that some rung hits. An average or almost-all version would not prove the conjecture, but would test whether the observed slow growth of record depth is explained by decorrelation rather than by a fixed bound. Identify precisely what replacement for the large sieve can accommodate the moving factorizations $q_r = p(p+r)/4$.

Problem 4. Determine whether $M(X) = \max_{p<X,\ p\in\mathcal{H}} D(p)$ is unbounded. Short of that, derive a tail estimate for $D(p)$ strong enough to distinguish a slowly diverging record from a fixed ceiling, and compare record setters with the smallest spectral margin $\min_{r<D(p)} |1 - \mathfrak{I}_r(p)|$ on their failed rungs. A bound $D(p) = o(p)$ for every sufficiently large p would already prove the conjecture on $\mathcal{H}$ after finite verification, so it is not a modest intermediate target.

Problem 5. Corollary 3.11 says which rungs *can* carry a level-S failure; it does not say which do. Below 10^{15} only $r = 51$ is occupied, though $r = 119$ and 123 are admissible and are visited by every ladder deeper than them — such ladders exist (the record is 155), and level RC now demonstrably reaches 123. Estimate the density of primes whose support subgroup at rung 119 or 123 equals the unique admissible subgroup there, and predict the height at which a second occupied rung should first appear. A prediction that survives one further decade would turn Corollary 3.11 from a constraint into a model.

9 Conclusion

The certificate hierarchy explains almost every failed rung in the hard class, usually through the Jacobi symbol alone. What remains is small in proportion but not in count: the residual share falls to 0.8227%, while 4.59 billion residual misses remain below 10^{15}. Those misses occur disproportionately at divisor-poor shifted values, yet their divisors are far too dependent modulo r for a random-occupancy argument. The one level that no real-character certificate detects is not scattered either: because $4A = p + r$ puts 4 in every support subgroup, level S can occur at only ten rungs below 300, and every observed instance sits at the smallest of them.

The depth data carries a separate message. The maximum first-hit rung stayed at 107 for a long interval and then rose through 131 to 155, with intermediate depths populated. This is the pattern expected from a very thin extreme tail. It provides no evidence for a uniform $O(1)$ depth bound. Our numerical expectation is instead that the record depth is unbounded but grows extraordinarily slowly; the census is far too short to identify that growth law.

None of this proves the conjecture. A vanishing share of residual failures does not prevent one prime from missing every rung, and an observed record is neither an upper bound nor a theorem of divergence. The useful outcome is the separation itself: local obstructions are finite and certifiable, while the remaining problem concerns the structured divisors of $p^2((p+r)/4)^2$ along shifted prime values. That is the part a proof must control.

Data and code availability

The complete public repository is `https://github.com/Attitude-Adjuster/erdos-strauss`. Questions, corrections, code problems, and independent reproduction reports may be submitted at `https://github.com/Attitude-Adjuster/erdos-strauss/issues`.

The repository contains the exact-arithmetic reference implementation, the independent C++ scanners (the frozen reference and an optimized variant proved output-identical against it), the CUDA scanner, the certificate verifier—including the mechanical check of Proposition 3.10—the per-shard summaries of both censuses (192 shards at 10^{13}, 400 at 10^{15}), the merged depth and level histograms, the 20,001 fine-certificate and deep-prime records of the 10^{13} census, the 585,677 of the 10^{15} census, and the verification transcripts. The analysis scripts reproduce the cumulative and disjoint-decade statistics. Level-J failures are summarized rather than stored individually because each is regenerated directly from (p, A, r).

Each solution certificate is a triple (p, A, u) and is checked through $4ABC = p(AB + BC + CA)$. The per-shard summaries are published so that small pieces of the census can be reproduced independently rather than requiring the full run.

Table 3: Numerical check of (2) against direct enumeration, reporting both the correct (conjugated) and the mis-transcribed (unconjugated) evaluations; the latter counts the class $(-q)^{-1}$ instead of $-q$.

p	A	r	rung status	(2)	unconjugated	direct
73	19	3	fails (J)	0	0	0
73	20	7	hits	8	6	8
73	21	11	hits	4	2	4
73	23	19	fails (J)	0	0	0
1129	285	11	hits	8	8	8
21169	5293	3	fails (J)	0	0	0
21169	5300	31	hits	6	6	6

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A detailed critical reading identified the composite-r gap in the original obstruction statement and sharpened the discussion of the analytic obstacles. It also supplied the independent divisor counts used in Table 3.

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